

DevOps Task - EC2 Access Control Sync & Repo Update

Objective

Harden and automate network access to a public web server on AWS while preserving open SSH access for emergency use.

Current State

- **Compute:** AWS public EC2 instance
- **Public IP:** 34.254.140.195
- **Domain:** 2bcloud.io (should resolve to the EC2 public IP from your PC/laptop)
- **Ingress today:** Security Group allows **SSH (22/tcp) from 0.0.0.0/0 only**
- **No WAF / ALB / reverse proxy** in front

Credentials

Linux user `credentials` will be `sent by email`.

YAML Template:

```
name: security-group
rules:
  ssh:
    - 0.0.0.0/0
  http:
    - 212.199.143.34/32
```

Requirements

1. **Script** (choose Shell or Python) that:
 - a. Your current **home/public IP** is a **constant** (CIDR /32).
 - b. The script must **not use hard-coded values for the region's name or security group**; both should be determined **dynamically** at runtime.
 - c. Fetches the **current Cloudflare IPv4 ranges**.
 - d. Reads the **YAML template** above as input state.
 - e. **Syncs the EC2 Security Group** so that:
 - i. **SSH (22/tcp)** remains **open to 0.0.0.0/0** (unchanged).
 - ii. **HTTP (80/tcp)** is **allowed from**:
 1. your current **home IP** address
 2. **all Cloudflare IP ranges**
 - iii. Remove any **stale** HTTP CIDRs not in {home IP, Cloudflare ranges}.
 - f. **Updates the YAML template** to reflect the **post-sync** rule set (sorted, de-duplicated).
 - g. Pulls/Clones, commits and **pushes the updated YAML to your public Git repository**.
2. **DNS**
 - a. Ensure 2bcloud.io records resolved to **52.215.116.12** so you can reach **from your PC/Laptop only!!** the server via <http://2bcloud.io>

Deliverables

- **A public Git repo** containing:
 - The **script**.
 - The **updated YAML** reflects the synced rules.
- **EC2 Security Group state**:
 - SSH (22) **open to 0.0.0.0/0**.
 - HTTP (80) **open to your current home IP /32 and all Cloudflare IP ranges**.
- **Functional test**: You can load the site from your local machine at <http://2bcloud.io>.

Acceptance Criteria

- Running the script is **idempotent** (re-runs don't create duplicate rules).
- Script logs: detected home IP, number of Cloudflare ranges, planned adds/removes, final SG rule count.
- Updated YAML exactly mirrors the applied HTTP sources (home IP + Cloudflare ranges).
- `curl -I http://2bcloud.io` from your local machine returns HTTP 200 (or the site's expected status).
- SSH remains reachable from the public internet on port 22.

After completion, please send a link to your public Git repository and the SSH credentials to **evgenyl@2bcloud.io** and **nof@2bcloud.io**.